

Symbolic validity of numerical NeuronBench recoveries

Post-hoc analysis of evolution runs 313195 and 313196

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Bottom line. Among 45 held-out fits, 29 attained $\text{NRMSE} \leq 10^{-6}$. The project’s SRBench-style symbolic checker labels 26 of these 29 as ground-truth matches. None is literally identical when its fitted decimal constants are treated as exact numbers, but every recovered frontier contains a polynomial with the correct ground-truth monomial support. Four best equations contain additional, numerically negligible monomials.

Question and method

The fully observable target is the current-balance law

$$f = I_{\text{ext}} - 120\phi_{\text{Na}}(V - 50) - 36\phi_{\text{K}}(V + 77) - 0.3(V + 54.4) - g_q\phi_q(V - E_q),$$

where the final channel depends on the neuron world (and is absent for `na_fatigue`). PySR fits f/s , where s is the training-target RMS. For this analysis each saved equation was remapped from x_i to the corresponding physical feature, multiplied by s , expanded with SymPy, and compared with the manifest ground truth.

Three notions must be kept separate:

- **Numerical recovery:** the existing held-out criterion, $\text{NRMSE} \leq 10^{-6}$.
- **Project symbolic match:** the same `evaluation.check_pysr_symbolic_match` logic used for SRBench and `evolve_pysr.py`. It rounds floating constants to three decimals and accepts a zero or constant difference, or a constant ratio. This is deliberately looser than literal algebraic equality.
- **Strict equality/support:** respectively, simplify $(\hat{f} - f) = 0$ with the stored decimals, and equality of the monomial sets after polynomial expansion. An exact target equation must itself pass the numerical threshold, so restricting the symbolic sweep to recovered frontier rows cannot discard an exact match.

Results

Evolution run	Numeric recoveries	Project match	Strict equality	Best support correct
313196 (trained on one world)	14	14	0	11
313195 (trained on two worlds)	15	12	0	14
Total	29	26	0	25

Of the 26 project matches, 23 have zero rounded symbolic difference and three have a constant rounded difference; none relies only on the constant-ratio rule. All 29 recovered frontiers contain at least one row with exactly the same monomial support as the physical ground truth. Thus the numerical recovery counts mostly represent recovery of the correct current-balance polynomial, with constants fitted to finite precision, rather than unrelated interpolating formulas.

Examples where “recovered” does not mean literally identical

1. A tiny extra nonlinear term. Run 313196, `na_fatigue`, seed 10000 has $\text{NRMSE} = 9.06 \times 10^{-11}$. Its expanded physical equation differs from ground truth by

$$\begin{aligned} \hat{f} - f = & -5.54 \times 10^{-7} \phi_{\text{Na}}^2 + 1.10 \times 10^{-6} \phi_{\text{Na}} - 9.19 \times 10^{-7} \phi_{\text{K}} - 1.07 \times 10^{-7} \\ & + \text{linear and bilinear corrections of magnitude at most } 3.3 \times 10^{-8}. \end{aligned}$$

The ϕ_{Na}^2 term makes the equation structurally different. The project checker still calls it a match because these tiny coefficients disappear under its rounding.

2. Correct support, approximate constants, and a project-checker miss. Run 313195, `ca_rebound`, seed 10000 has $\text{NRMSE} = 3.09 \times 10^{-7}$. The relevant physical coefficients are:

Term	Ground truth	Found	Term	Ground truth	Found
I_{ext}	1	0.99999762	$V\phi_{\text{Na}}$	-120	-119.99999734
$V\phi_{\text{K}}$	-36	-35.99997305	$V\phi_{\text{T}}$	-3.2	-3.20012909
V	-0.3	-0.29995093	ϕ_{Na}	6000	5999.99985004
ϕ_{K}	-2772	-2771.99906434	ϕ_{T}	384	383.99682787
1	-16.32	-16.31889694			

This is the same polynomial support with very close coefficients, yet the project checker returns no match. Its three-decimal rounding is applied to the normalized, factorized PySR expression before expansion, so the decision can depend on how an otherwise similar polynomial is factored.

3. An even smaller checker false negative. Run 313195, `d_type`, seed 10004 has $\text{NRMSE} = 4.30 \times 10^{-10}$ and the correct support. For example, the ground-truth coefficients of $V\phi_{\text{D}}$, ϕ_{D} , and the constant are -9 , -693 , and -16.32 ; the fitted values are -8.999999917 , -692.999995373 , and -16.320003695 . It also fails the project checker despite being an excellent coefficient-level recovery.

Interpretation

Yes, there are $\text{NRMSE} < 10^{-6}$ equations that are different from ground truth. In the strict stored-decimal sense this is true for all 29 recoveries, which is expected when continuous constants are optimized numerically after target scaling. The more substantive structural difference occurs in four best equations, all through tiny extra monomials. Conversely, three numerically recovered equations with correct support are missed by the current project checker.

For future NeuronBench summaries, the most informative reporting is therefore a three-part label: held-out numerical recovery, the existing SRBench-style match, and canonical expanded-polynomial support (optionally with a stated coefficient tolerance). This avoids treating fitted-decimal inequality as a scientific miss while making small spurious terms visible.